

LaserLockAI

Official Operator, User Manual

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Chapter 1: System Overview & Architecture

LaserLockAI is a closed-loop simulation and testing platform built for high-precision optical pointing, acquisition, and tracking (PAT) of laser beacons. It operates deterministically at a 30 Hz step rate.

Core Subsystems:

Discrete Physical Engine: Computes 6-DOF relative kinematics and line-of-sight geometry.

FPA Optical Camera Sensor: Synthesizes 640x480 monochrome frames displaying Gaussian beam irradiance.

Disturbance Generator: Injects atmospheric extinction, mechanical jitter, and multiple noise profiles.

Hybrid Detection Engine: Evaluates candidate optical spots using Classical Computer Vision (CV) or a YOLO AI Neural Detector.

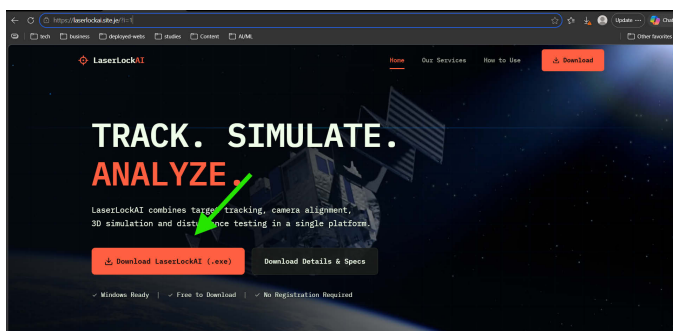
Kalman Kinematic Filter: Smooths measurement jitter and estimates velocity vectors.

2-Axis PTZ Gimbal Controller:

Drives the optical error toward zero using dual-channel PID loops with anti-windup clamping.

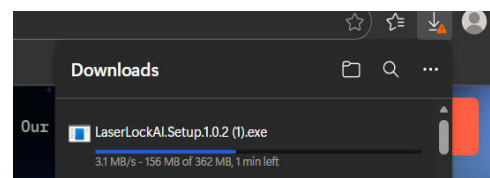
Compliance Targets (Problem Statement 4): Acquisition Time: ≤ 2.0 seconds. Tracking Error: ≤ 10.0 pixels. Target Loss Rate: $< 5.0\%$ of the duration. Re-acquisition Time: ≤ 1.0 second. Processing Speed: ≥ 20.0 FPS.

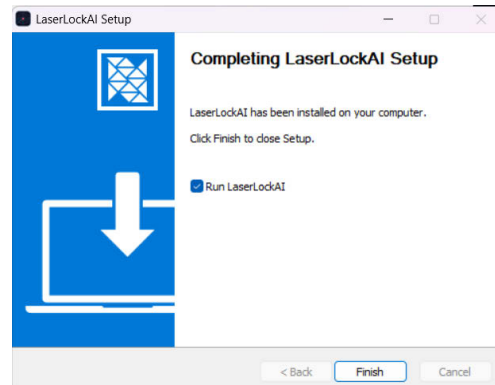
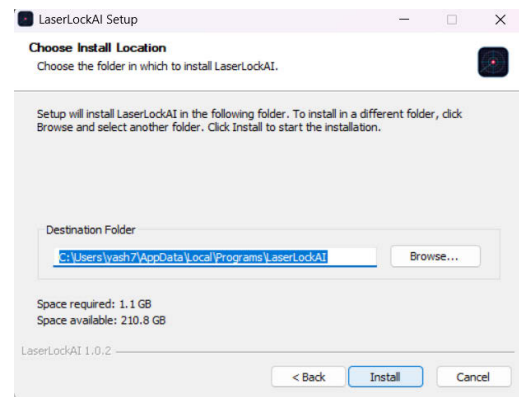
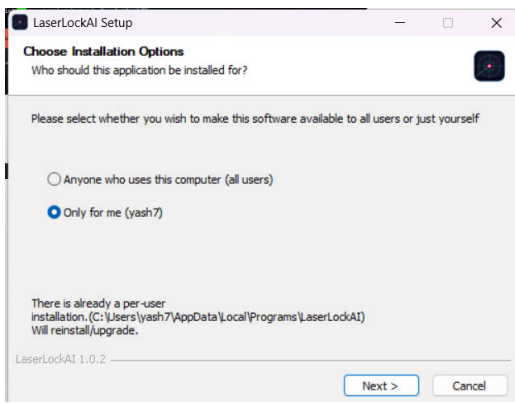
Chapter 2: Installation & Quick Start Guide



1-Open the link- <https://laserlockai.site.je/>

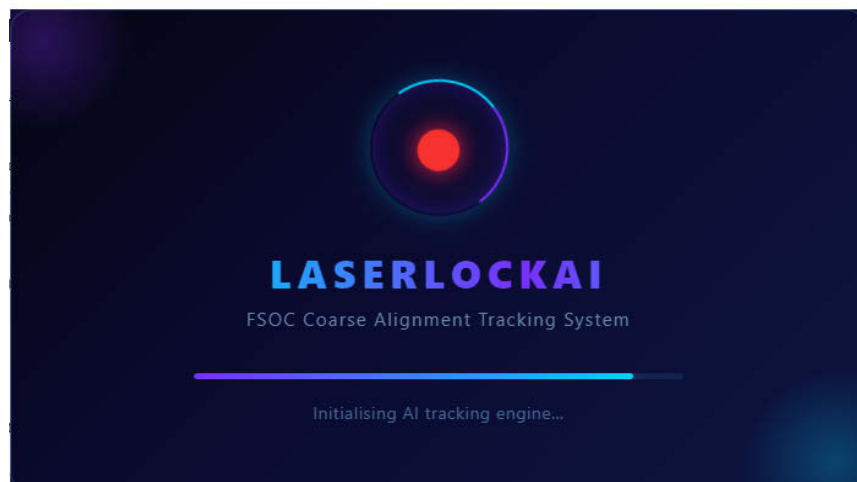
2-Click on Download LaserLockAI(.exe) as shown with green arrow in the image, this should start downloading an .exe file.



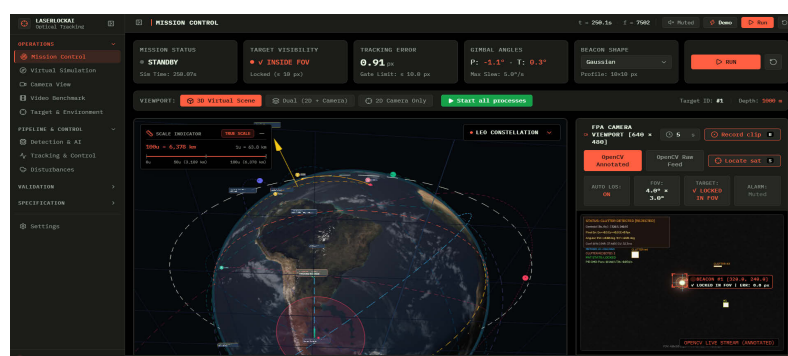


After the download click next and then click install to install the software in the selected directory. After the installation is done click finish and run the software.

The startup should look like this :



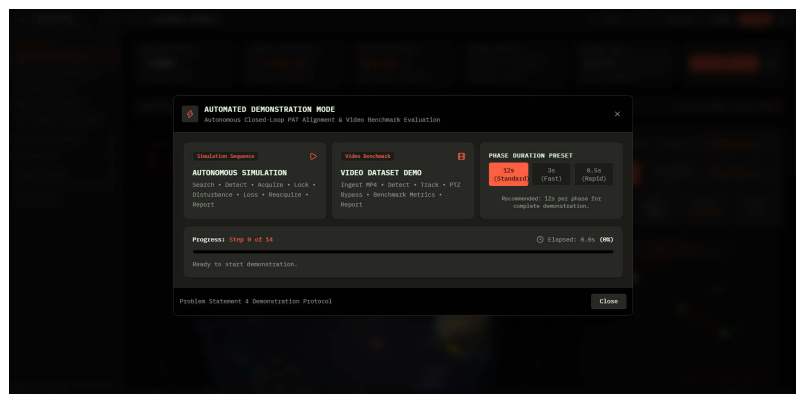
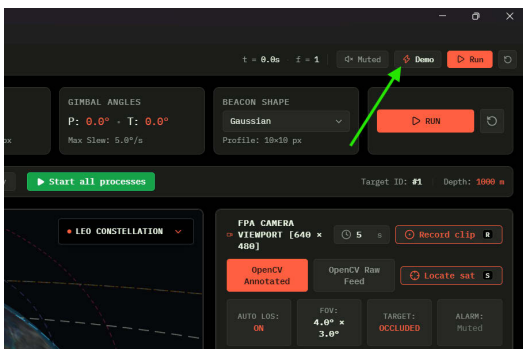
The first page will open after the startup :





Chapter 3: Auto-Demonstration Mode

The automated orchestrator allows rapid evaluation without manual tuning. Access it by clicking the orange "Demo" button in the top navigation header.

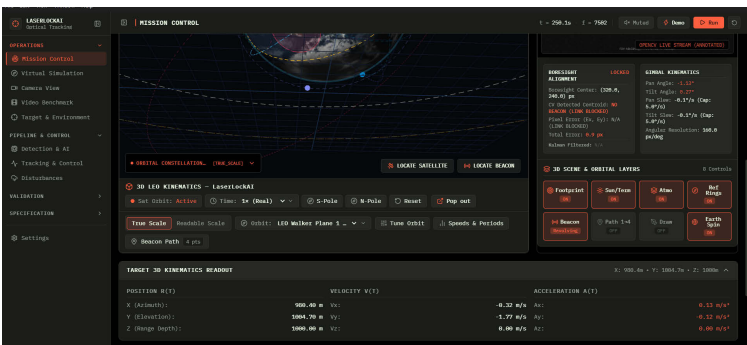


Demonstration Modes:

This interface will open after the click.

- **Simulation Demo:** Triggers the physical engine to demonstrate target illumination, circular orbit kinematics, and closed-loop PID gimbal tracking.
- **Video Benchmark Demo:** Loads pre-recorded MP4 footage into the evaluator to demonstrate centroid annotation and frame-by-frame error scoring.
- **Interactive Guided Walkthrough:** Provides a step-by-step UI tour highlighting key sensors and telemetry widgets.

Chapter 4: Mission Control Operations



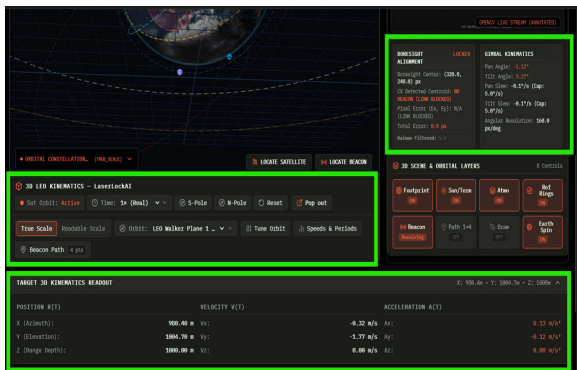
Mission Control acts as the primary tactical dashboard, merging 3D orientation, live camera feeds, and real-time kinematic metrics.

Primary Viewports:

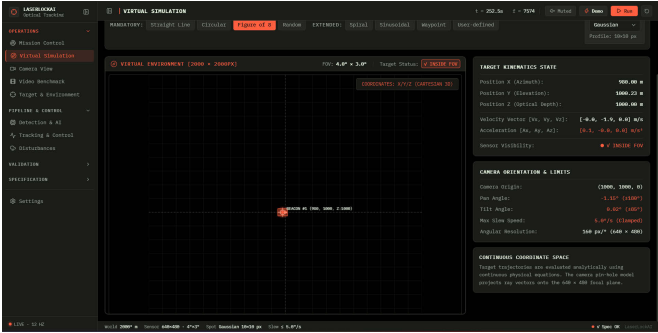
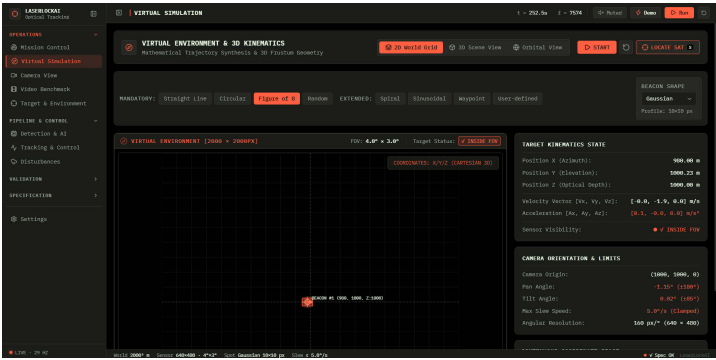
- 3D Scene Visualizer: Displays the host satellite, line-of-sight vector, and target trajectory trails in Euclidean space.
- FPA Camera Viewport: Shows the live 640x480 simulated sensor feed, including bounding boxes and centroid markers
- Pointing Error Polar Scope: Visualizes instantaneous angular error relative to the camera boresight via concentric rings.

Key Telemetry Indicators:

- PAT State Machine: Indicates current phase (IDLE, ACQUIRING, TRACKING, or LOST).
- Tracking Error: Reports the absolute distance in pixels between the beam centroid and the optical origin.
- Gimbal Slew Rates: Displays live pan and tilt angular velocities.
- Signal-to-Noise Ratio (SNR): Compares peak beacon power to the background noise floor.



Chapter 5: 3D Virtual Simulation & Orbits



2D(Birds eye view)

This module provides both a local physical testbench and a macroscopic Earth orbital mechanics engine.

- 2000 × 2000 Virtual World: Features a calibrated 2000 × 2000 pixel/meter Cartesian coordinate space with boundary reflection dynamics.
- Optical Projection Frustum: Visualizes the 640 × 480 FPA camera's 4.0° × 3.0° field-of-view (FOV) projection cone (160 px/deg) originating from the 2-axis PTZ gimbal.
- Target Kinematic Trajectories: Supports all 4 mandatory motion generators (Straight Line at 150 px/s, Circular, Figure of 8, and Random uniform walk).
- Optical Beacon Spot Profiles: Emits configurable optical spots including 10 × 10 px Square (PS4 Baseline), Circle, and Gaussian intensity distributions.
- Real-Time Gimbal Tracking: Renders instantaneous boresight crosshairs, target lead vectors, and coarse pointing error convergence in real time.

Chapter 6: Camera View & Gimbal Operations

This page is dedicated to payload monitoring and manual actuator overrides. Camera & Gimbal Specs:

- Field of View: 4.0° Horizontal x 3.0° Vertical with a center origin at (320.0, 240.0) px.
- Rotation Bounds: Pan operates from -180° to +180°; Tilt operates from -90°to +90°
- 3-Velocity Clamping: Physical actuators are strictly governed at ≤ 5.0°/s to prevent motor saturation.

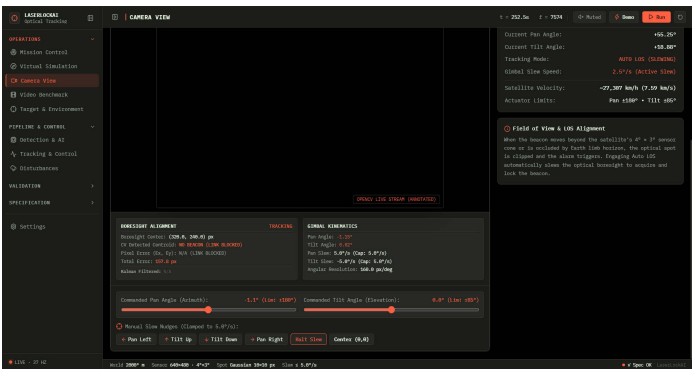
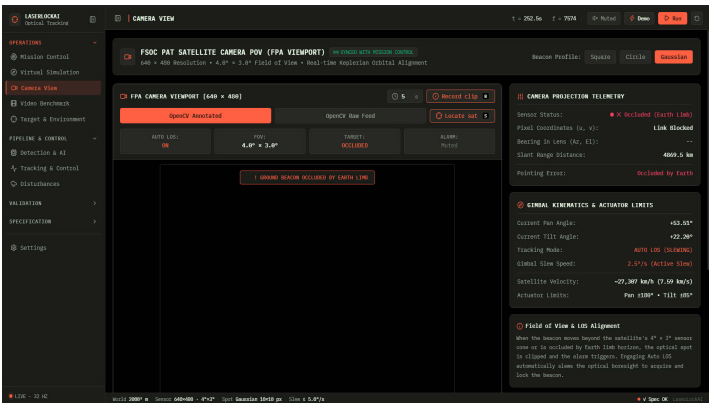
Manual Operations:

Nudging: Operators can inject intentional

- angular offsets (e.g., +1.0° Pan) to observe closed-loop compensation.

Zero Boresight: Instantly commands

- the actuators to return to their 0.0° center alignment.



Chapter 7: Video Benchmark & Samples

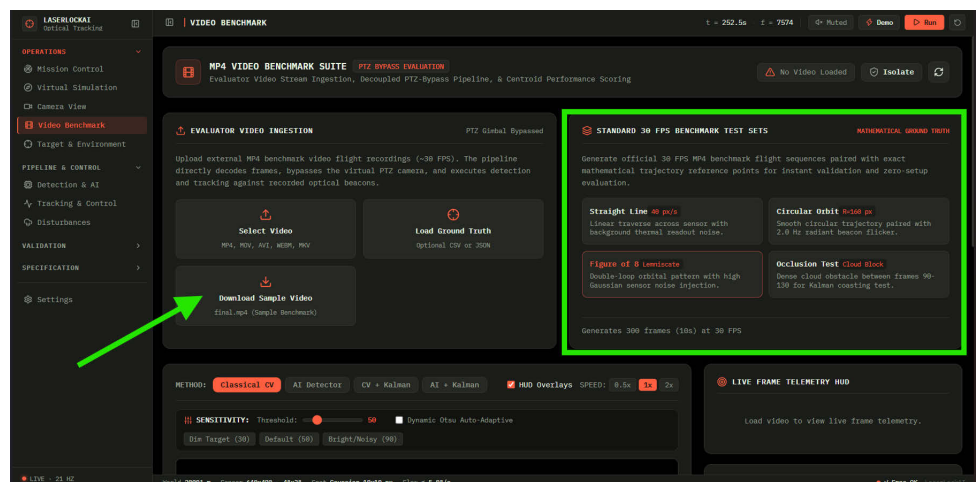
The benchmark evaluator allows direct algorithm testing against pre-recorded flight videos by bypassing the PTZ physical movement.

Synthetic Test Sets:

- Straight Line: Linear target traverse paired with thermal readout noise.
- Circular Orbit: Harmonic trajectory with a 2.0 Hz beacon flicker modulation.
- Figure of 8: Complex multi-axis aerial maneuvers with Gaussian noise.
- Occlusion Test: Dense cloud obstruction inserted between frames 90-100 to validate Kalman coasting.

Click on "Download Sample Video"
Videos can be downloaded locally or
directly mounted into the evaluator.

Then click on select video to select
the downloaded sample video from
your device.

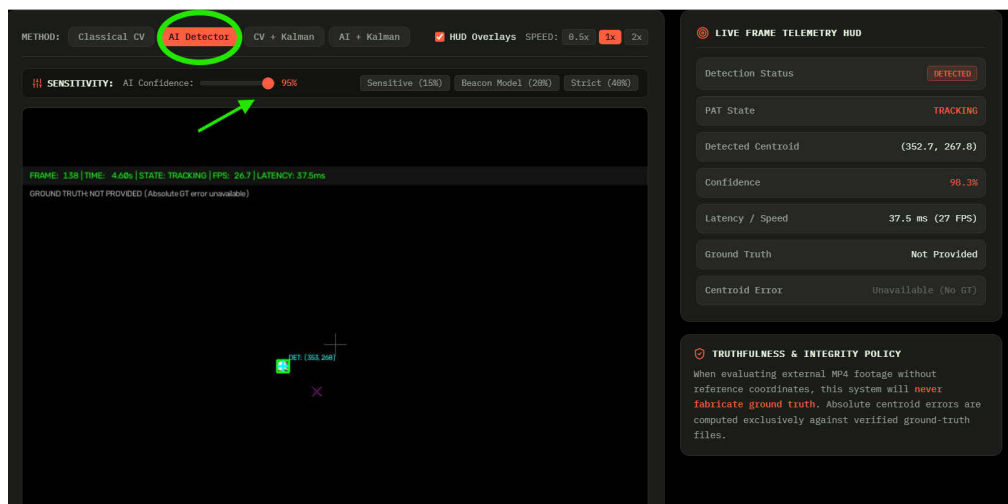


After the video is selected Click on AI
Detector or AI + Kalman.

Keep the AI Confidence to 95%(max.)

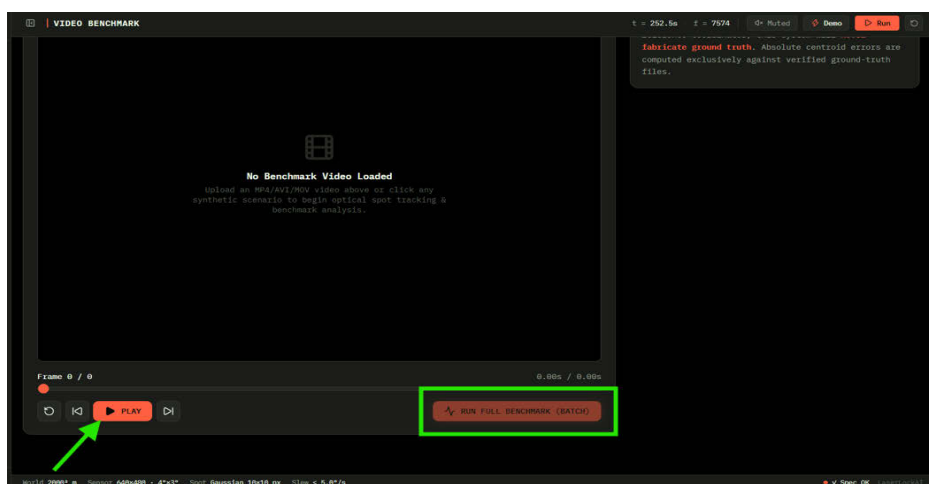
In the LIVE FRAME TELEMETRY
HUD(located far right in the
Interface).

you can see how the AI model is
performing in real time on the given
video.



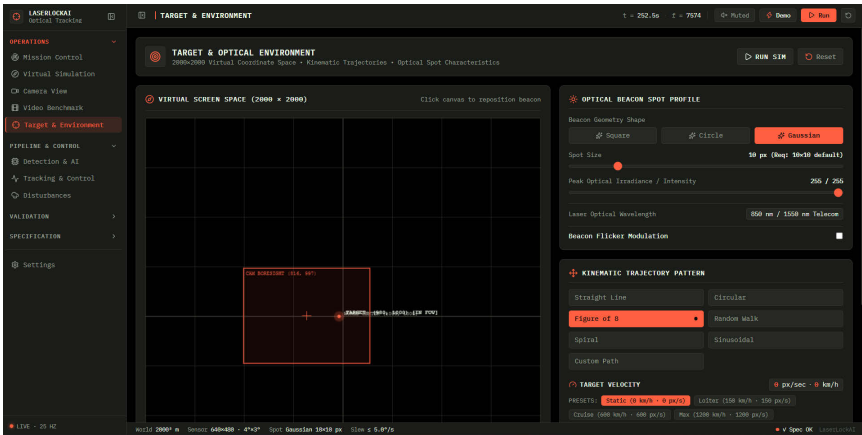
Here clicking on Play you can
see the algorithm detecting
the target in the video live.

In the bottom right you can
see a button "RUN FULL
BENCHMARK (BATCH)", using
which you can ...evaluate the
full video at high speed,
calculate key accuracy
metrics, and export
performance reports.



Chapter 8: Target & Optical Beacon Setup

Operators can completely customize the optical target's geometric and dynamic properties.



Target Configuration:

- Shape Models: Select between Square (10x10 px), Circle, or a Gaussian Spot.
- Irradiance Profile: Adjust the peak intensity (0-255) and beam divergence radius for realistic laser simulation
- Flicker Modulation: Simulate pulsed transmitters with frequencies ranging from 0.5 Hz to 10.0 Hz.

Kinematic Trajectories:

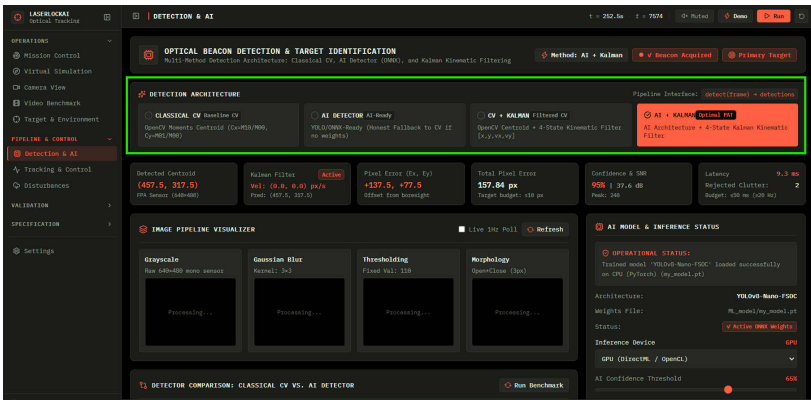
Configure the target to execute a Straight Line, Circular Orbit, Figure of 8, or a stochastic Random Walk

Chapter 9: Detection Engine (CV & AI)

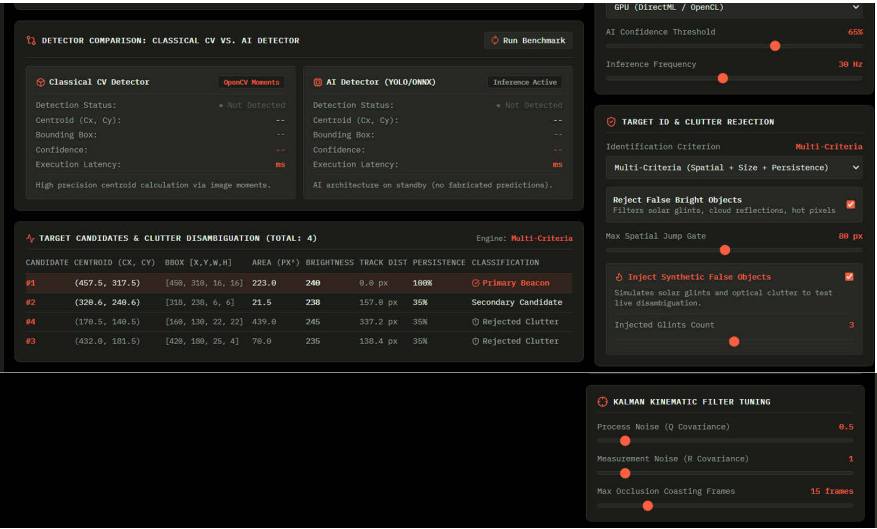
The system supports four distinct operational pipelines for target identification.

Detection Architectures:

-Classical CV (Moments): Uses adaptive thresholding and spatial image moments to compute sub-pixel coordinates at ultra-low latency.



- AI Detector (YOLO): Deploys a deep neural network to draw bounding boxes and reject solar glint.
- CV + Kalman: Combines classical spatial moments with predictive filtering.
- AI + Kalman (Full Hybrid): Merges YOLO neural inference with kinematic state estimation for state-of-the-art precision.



Tuning Controls:

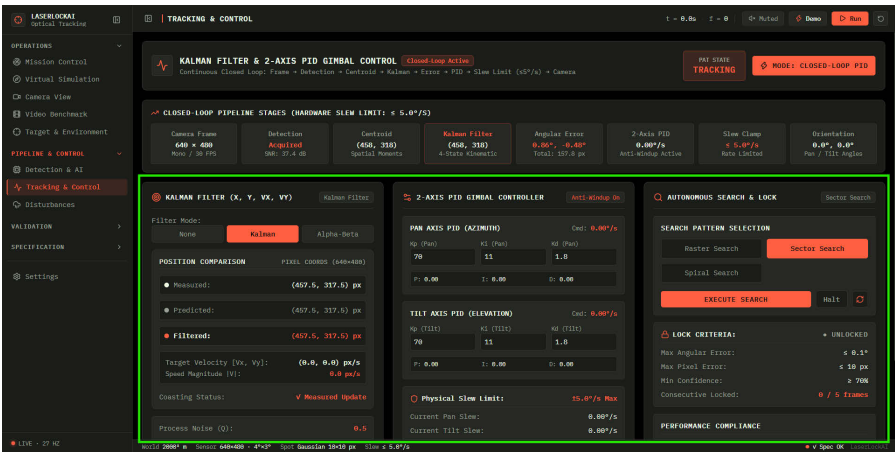
Intensity Threshold: Filters dark background pixels (adjustable from 0-255).

Dynamic Otsu: Automatically calculates the optimal binarization threshold.

Contour Area Bounds: Rejects objects too large (solar reflections) or too small (hot pixels)

Chapter 10: Tracking & 2-Axis PID Gimbal

The closed-loop tracking system links the detection centroid outputs directly to the mechanical actuators.



Kalman Kinematic Filter:

1-State Estimation: Tracks position and velocity vectors while filtering measurement noise.

2-Dead-Reckoning Coasting: Extrapolates target trajectory for up to 15 frames during sudden cloud occlusions to prevent gimbal runaway.

PID Gimbal Controller:

-Proportional (Kp): Drives the gimbal toward the target based on immediate centroid offset.

-Integral (Ki): Eliminates persistent steady-state tracking errors.

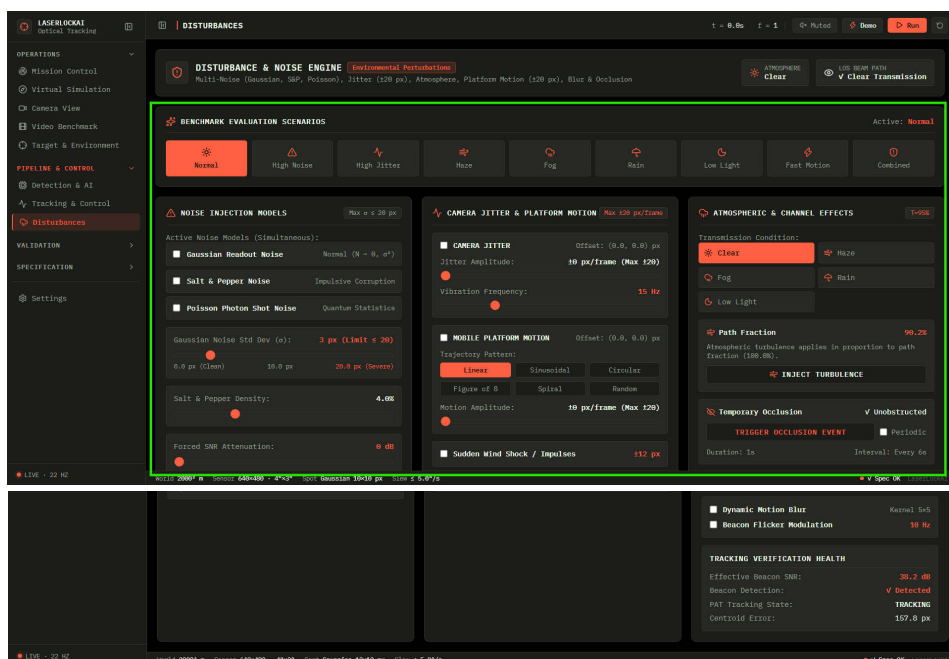
-Derivative (Kd): Damps the velocity of the actuators to prevent overshoot. Anti-Windup: Integrator accumulation is clamped to prevent windup lag.

Autonomous Search & Lock:

- Search Scan Patterns: Autonomous Raster, Sector, and Spiral search routines to sweep the gimbal and find beacons outside the camera FOV.
- Coarse Lock Verification: Locks tracking when centroid error is maintained within ≤ 10 pixels for consecutive frames.
- Rapid Re-Acquisition: Recovers optical lock within ≤ 1.0 second following temporary line-of-sight dropouts.

Chapter 11: Disturbances & Environmental Stress

This module tests the tracking pipeline against harsh external variables.



Noise & Jitter Generators:

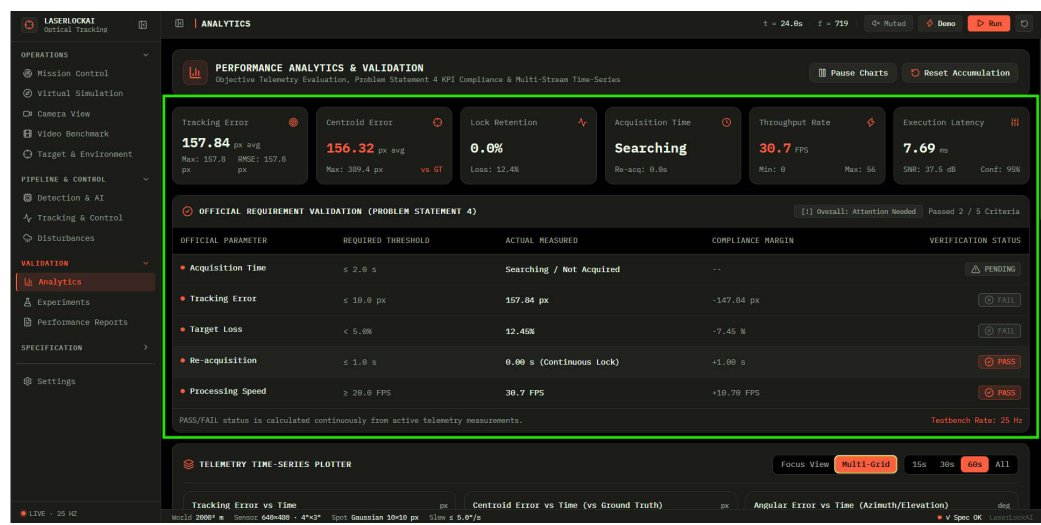
- Gaussian Noise: Simulates thermal readout amplifier interference.
- Salt & Pepper Noise: Simulates dead pixels and high-altitude radiation hits.
- Poisson Noise: Models quantum photon arrival statistics.
- Camera Jitter: Injects high-frequency mechanical vibrations mimicking reaction wheels or engines
- Continuously scores Acquisition Time, Tracking Error, Target Loss, Re-acquisition Time, and Processing Speed
- The system awards PASS/FAIL badges in real time based on benchmark margins.

Atmospheric Channels:

- Toggle between Clear Sky, Haze, Dense Fog, or Rain to alter optical transmission and scattering.
- Occlusion Trigger: Instantly drops a dense obstacle into the flight path to validate target re-acquisition

Chapter 12: Analytics & Real-Time Telemetry

- The analytics engine strictly evaluates the active flight test against the formal SIH requirements.

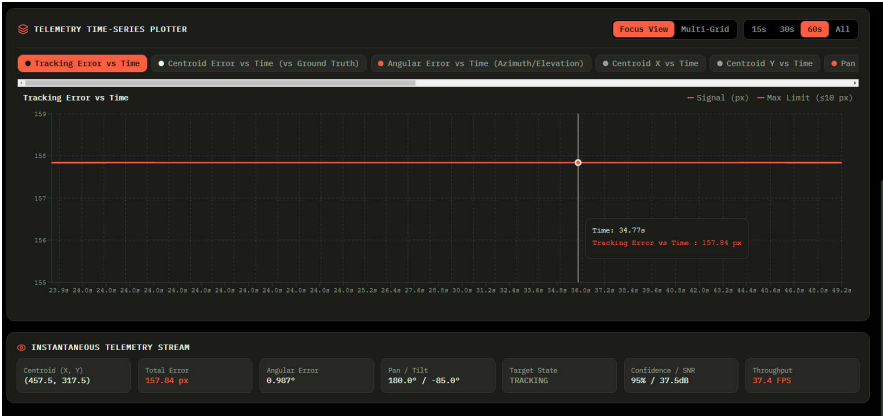
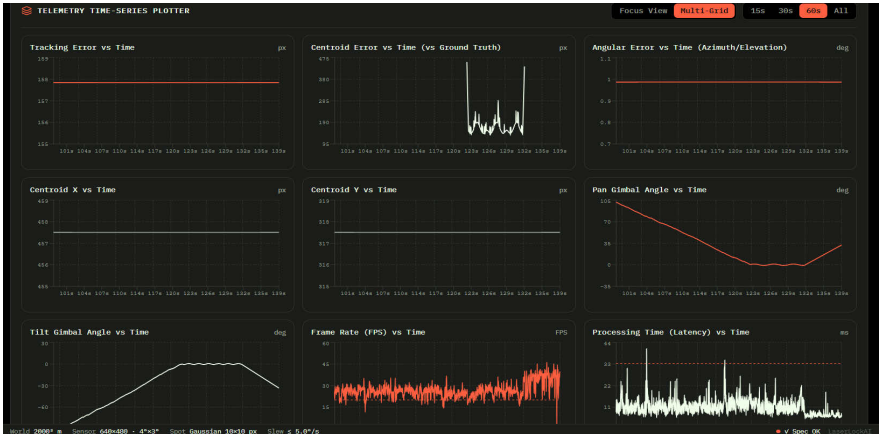


Validation Engine:

- Continuously scores Acquisition Time, Tracking Error, Target Loss, Re-acquisition Time, and Processing Speed.
- The system awards PASS/FAIL badges in real time based on benchmark margins.

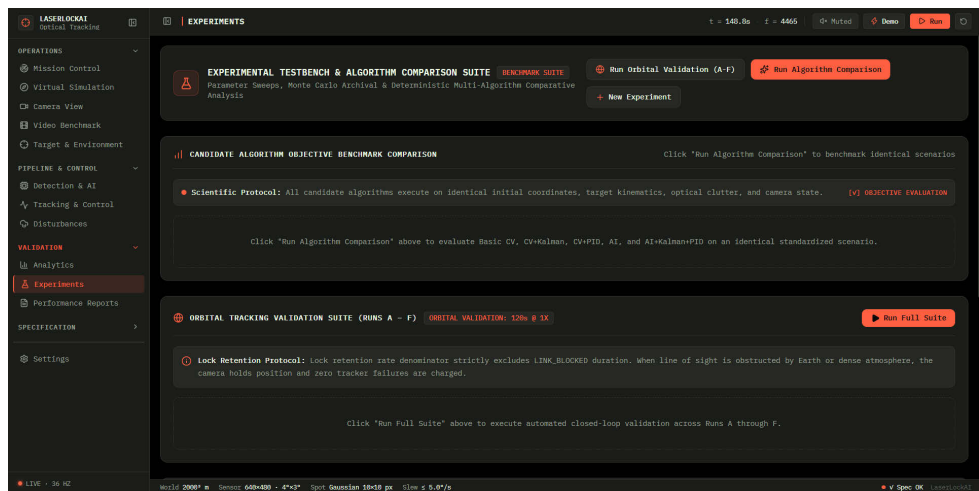
Live Charting:

- 11 independent telemetry streams track Pointing Error, Centroid Coordinates, Slew Rates, and FPS Latency.
- Operators can adjust the rolling history window (50, 100, 200, or 500 frames) for macro or micro analysis.



Chapter 13: Experiments & 5-Algorithm Benchmarks

An automated laboratory designed to run side-by-side comparative testing.



The screenshot shows the 'ARCHIVED EXPERIMENTAL TESTBENCH TRIALS (6)' section. It includes buttons for 'Export CSV' and 'Export JSON'. Below is a table with the following data:

EXPERIMENT ID	TIME	TRIAL NAME	ALGORITHM	TRAJECTORY	DISTURBANCE	MEAN ERROR	THROUGHPUT	STATUS	ACTIONS
EXP-26261003-B23989	01:39 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (3px)	1.54 px	37.3 FPS	PASS	
EXP-26261003-B0A1E7	01:27 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (5px)	126.25 px	55.8 FPS	FAIL	
EXP-26261003-6099CA	01:06 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (5px)	126.38 px	32.2 FPS	FAIL	
EXP-26260927-A7BCA6	05:35 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (5px)	126.37 px	17.9 FPS	FAIL	
EXP-26260927-00AA94	03:05 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (5px)	126.34 px	32.1 FPS	FAIL	
EXP-26260927-61AF9F	02:55 PM	Live Verification Trial	AI + Kalman	Figure of 8	Gaussian (5px)	126.35 px	16.5 FPS	FAIL	

Benchmarking Protocol:

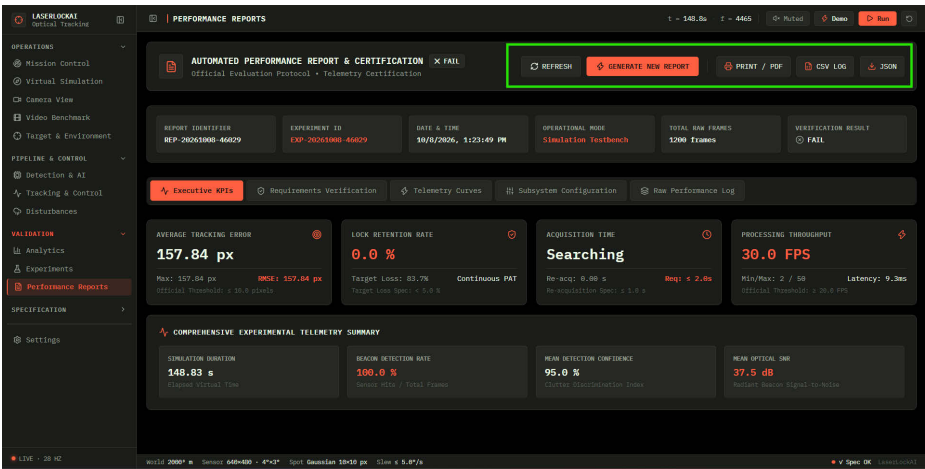
- Executes five distinct tracking configurations (Basic CV, CV+Kalman, CV+PID, AI Only, and AI +Kalman+PID) under identical flight profiles.
- Custom Trial Designer: Allows operators to specify trial duration, target motion, and noise profiles.
- All experimental runs are archived locally to JSON databases with unique experiment IDs for later review.

Chapter 14: Performance Reports & Exports

Generates auditable, formal documentation verifying flight metrics.

Click on Refresh to read and reload existing saved reports from the backend/disk.

Click on Generate new report to create a brand new remote immediately of the running simulation.



Export Contents:

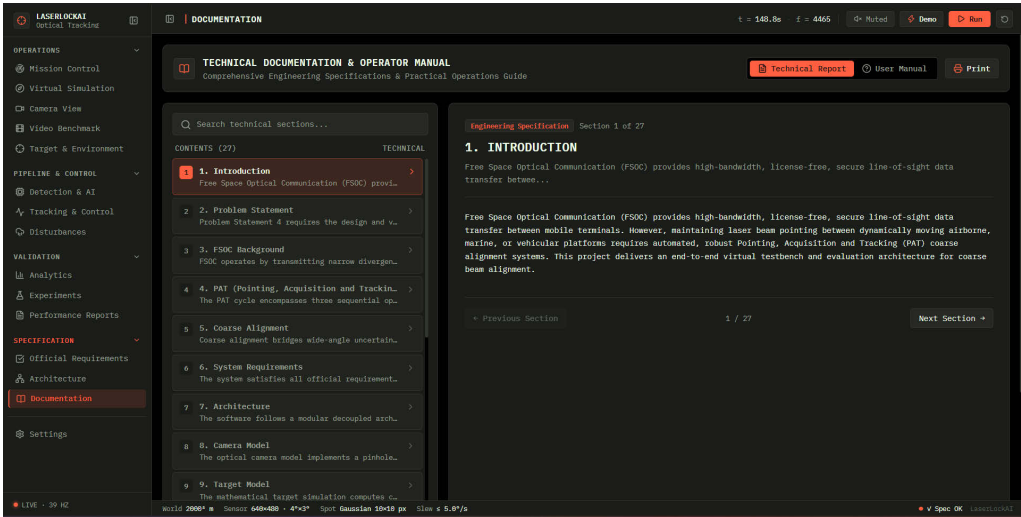
- Executive Summary: Outlines mission parameters, total frames, and overall pass rates.
- Error Statistics: Calculates Mean Error, RMSE, and standard deviation limits.
- Contextual Data: Logs the exact noise, weather, and jitter settings active during the evaluation.

Export Formats:

- PDF Report: High-fidelity, print-ready document containing all charts and compliance tables.
- CSV Raw Log: Frame-by-frame comma-separated values tracking angles, errors, and state changes.
- JSON Dataset: Structured arrays for deep programmatic analysis.

Chapter 15: Diagnostics & Troubleshooting

A field diagnostic reference for correcting common operational anomalies.



Common Remedies:

- Beacon Not Detected: Lower the intensity threshold or reduce atmospheric fog density
- Gimbal Oscillating/Hunting: Reduce the Proportional gain (K_p) and slightly increase Derivative damping (K_d).
- Steady-State Error Offset: Increase the Integral gain (K_i) to eliminate lingering tracking lag.
- Target Lost During Slew: Verify the target is not maneuvering faster than the $5.0^\circ/\text{s}$ mechanical limit. Ensure Kalman coasting is enabled.
- Low FPS/Latency Issues: Disable excess browser extensions, ensure hardware acceleration is active, and verify the backend is running properly on port 8000

-----END-----